Statistical Mechanics: Sound, Quantum Oscillators, and the Second Law

Leonard Susskind

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Chapter 1

Sound, Quantum Oscillators, and the Second Law

Overview. The second law is the destination, but two concrete calculations prepare the way. The speed of sound tests how thermodynamic derivatives become observable motion. The harmonic oscillator then exposes the failure of classical equipartition for stiff degrees of freedom and shows how quantum mechanics activates modes one by one. We finally return to the apparent conflict between reversible microscopic dynamics and increasing entropy, using phase-space flow, coarse-graining, chaos, and recurrence to state the second law as an overwhelmingly probable law rather than an exceptionless mechanical prohibition.

1.1 A Promise About the Second Law

The central problem is easy to state and difficult to explain. Newtonian equations are reversible: if a microscopic history is allowed, the time-reversed history is allowed as well. The second law nevertheless says that macroscopic processes have a preferred direction, characterized by entropy that ordinarily increases. We shall come to that tension, but not immediately. It is better first to see statistical mechanics produce concrete information and then encounter, within an elementary calculation, the historical puzzle that forced classical physics to give way to quantum physics.

The first example is numerical and deliberately rough: estimate the speed of sound in a dilute gas. The second is exact within its assumptions: calculate the mean energy of a harmonic oscillator in a heat bath. The oscillator will lead from equipartition to quantized energy levels, from one mode to infinitely many modes, and finally back to the question of what it means for a many-particle system to forget its initial state.

1.2 The Molecular Scale of the Speed of Sound

Imagine a small overdensity in a dilute gas. It spreads because molecules leave the crowded region, so its propagation speed should be comparable with the molecular speed. Equipartition gives the kinetic energy of one molecule,

$$\frac{3}{2}k_BT = \frac{1}{2}m\left\langle v^2\right\rangle, \qquad \left\langle v^2\right\rangle = \frac{3k_BT}{m}. \tag{1.1}$$

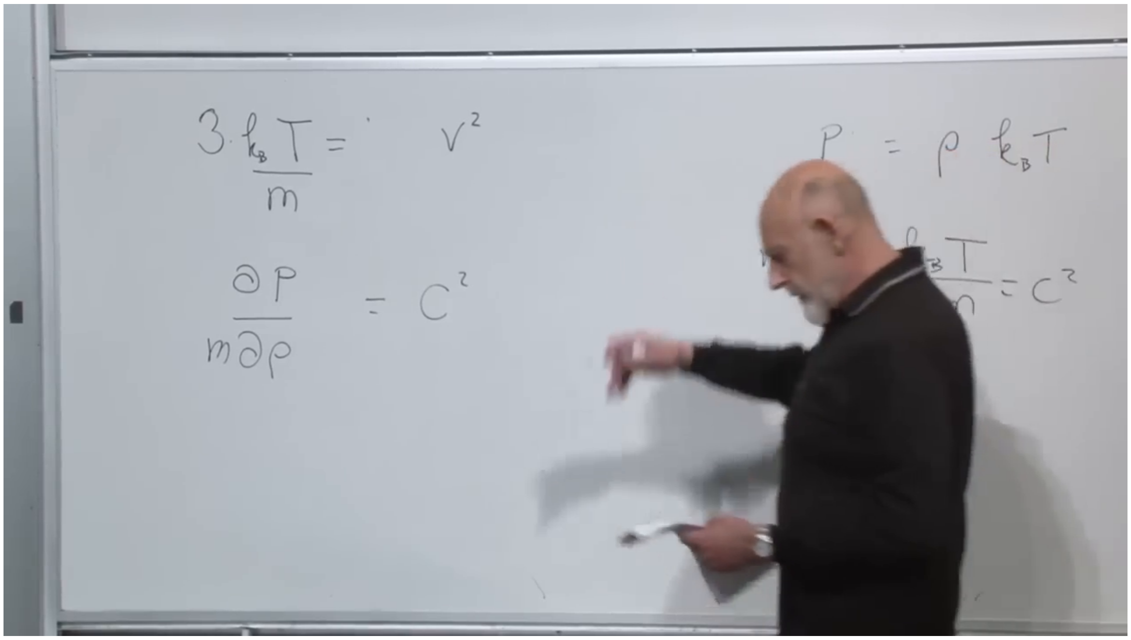

Figure 1.1: The molecular-speed estimate, the pressure derivative, and the ideal-gas substitution displayed together.

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Here temperature is measured from absolute zero and k_B has been restored because we want laboratory units. The argument is only a scale estimate: the molecular root-mean-square speed is not itself the speed of a collective sound wave.

A more systematic derivation considers a small element of gas. Unequal pressures on opposite faces exert a net force, and its acceleration is inversely proportional to the mass density. The result has the form

$$c_s^2 = \left(\frac{\partial P}{\partial \rho_m}\right)_{\mathcal{C}}, \qquad \rho_m = m\rho, \tag{1.2}$$

where ρ is number density and the subscript $\mathcal{C}$ specifies the thermodynamic condition followed by the disturbance. The board calculation uses

$$P = \rho k_B T \quad \Longrightarrow \quad \frac{1}{m}\left(\frac{\partial P}{\partial \rho}\right)_T = \frac{k_B T}{m}. \tag{1.3}$$

This is the isothermal sound-speed estimate. It differs from (1.1) by a factor of 3 in the squared speed, or $\sqrt{3}$ in the speed.

There is an important refinement. A small-amplitude sound wave normally compresses and rarefies the gas faster than heat can flow between neighboring elements. The local motion is therefore approximately adiabatic, not isothermal:

$$c_s^2 = \left(\frac{\partial P}{\partial \rho_m}\right)_S. \tag{1.4}$$

For an ideal gas with adiabatic index γ,

$$P\rho_m^{-\gamma} = \text{constant}, \qquad c_s^2 = \gamma \frac{P}{\rho_m} = \gamma \frac{k_B T}{m}. \tag{1.5}$$

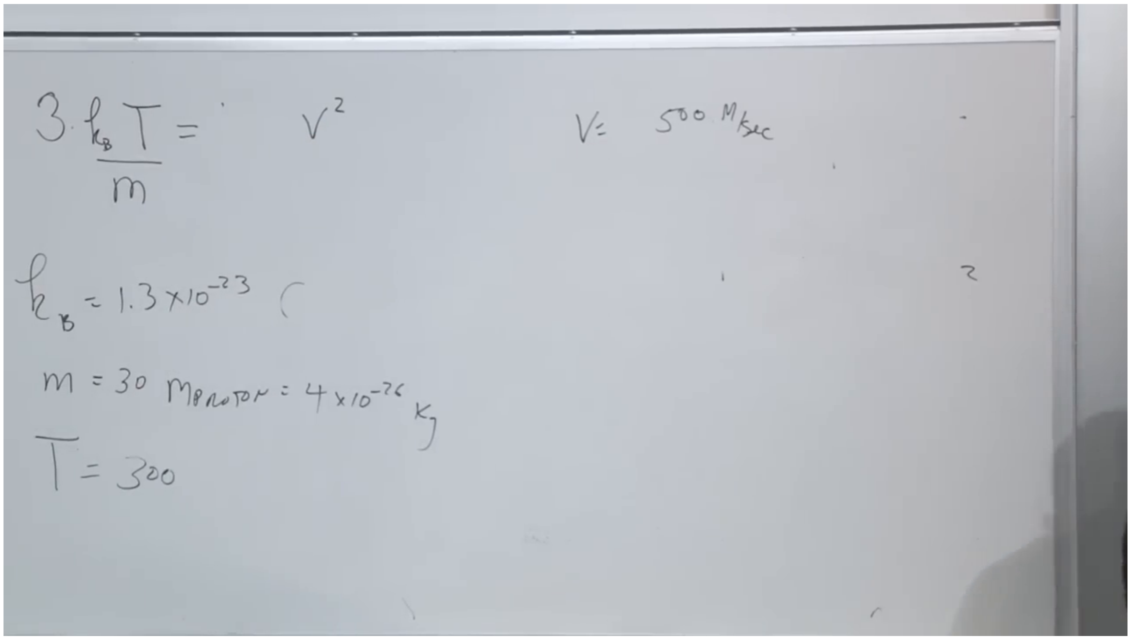

Figure 1.2: The room-air estimate using Boltzmann's constant, an effective molecular mass near 30 proton masses, and $T \simeq 300$ K.

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For room-temperature diatomic air, $\gamma \simeq 7/5$. Thus the fixed-temperature calculation is a useful first estimate, while the adiabatic derivative supplies the physically appropriate refinement.

Using the rough values

$$\begin{aligned} k_B &\simeq 1.3 \times 10^{-23}\ \mathrm{J\,K^{-1}}, \quad T \simeq 300\ \mathrm{K}, \\ m_{\text{air}} &\simeq 4 \times 10^{-26}\ \mathrm{kg}. \end{aligned} \tag{1.6}$$

the molecular estimate is about 500 $\mathrm{m\,s^{-1}}$. Temperature here must be an absolute temperature in kelvins, not an offset scale such as Fahrenheit. The familiar sound speed is of order 300 $\mathrm{m\,s^{-1}}$, and the adiabatic ideal-gas formula gives about 340 $\mathrm{m\,s^{-1}}$ at room temperature. The rough calculation has therefore found the correct physical scale.

Long before a microscopic account was available, one could determine the sound speed from measurements of pressure as a function of density. The statistical argument adds the striking interpretation that the resulting macroscopic speed is set by the thermal speed of individual molecules.

Question from the audience. Should neutrons be added to the molecular mass, and does the number 14 for nitrogen count only protons?

Response. The quoted atomic weight already counts protons and neutrons. Molecular nitrogen contains two atoms of atomic weight about 14, so its mass is about 28 proton masses; rounding 28 to 30 is sufficient for this estimate. [1]

[1] Lecture timestamp: 00:10:58–00:11:16.

Question from the audience. The factor 3 came from three translational directions, whereas sound propagates along one direction. Should that remove the factor of 3?

Response. It correctly warns us not to identify the three-dimensional root-mean-square molecular speed with a one-dimensional propagation speed. The opening argument was only an order-of-magnitude picture. A controlled answer comes from the pressure derivative, together with the thermodynamic condition appropriate to the wave. [2]

Question from the audience. Did the derivative assume that temperature remains constant during the wave?

Response. The displayed ideal-gas substitution did. Holding temperature fixed is a plausible provisional estimate for a very small disturbance; the standard refinement is that ordinary sound is approximately adiabatic, which replaces the isothermal derivative by $(\partial P/\partial \rho_m)_S$. [3]

Question from the audience. If the pressure and density variations are both small, does that make their derivative small?

Response. No. A derivative is the ratio of two small variations in the infinitesimal limit. Both δP and $\delta\rho$ may be small while $\delta P/\delta\rho$ remains finite and sets the wave speed. [4]

Question from the audience. Should the speed of sound rise when the gas is compressed, as it does when one passes from a gas toward a liquid or solid?

Response. Yes. The derivative is evaluated at the state under consideration, and $P(\rho, T)$ need not remain linear. When molecules are squeezed close together, the equation of state becomes nonideal and $\partial P/\partial\rho$ can grow substantially. [5]

Question from the audience. Can the value 500 $\mathrm{m\,s^{-1}}$ be used at every pressure?

Response. No. It belongs to the low-density range in which the ideal-gas equation is a useful approximation. Once higher powers of density enter $P(\rho, T)$, the derivative and therefore the sound speed depend on the pressure and density. [6]

Question from the audience. Does the original picture of sound spreading at roughly the molecular speed also fail as density increases?

Response. Probably. That picture is tailored to a dilute gas. Collective interactions become increasingly important at high density, which is precisely why the equation-of-state derivative is the more general description. [7]

[2] Lecture timestamp: 00:11:12–00:12:00.
[3] Lecture timestamp: 00:12:59–00:14:01.
[4] Lecture timestamp: 00:14:03–00:14:23.
[5] Lecture timestamp: 00:14:25–00:15:13.
[6] Lecture timestamp: 00:15:14–00:16:29.
[7] Lecture timestamp: 00:16:32–00:16:53.

Question from the audience. Why was the nitrogen-molecule mass written as roughly 30 proton masses rather than 28?

Response. Only numerical accuracy appropriate to the estimate is needed, so 28 was rounded to 30. Dividing the original 500 $\mathrm{m\,s^{-1}}$ estimate by $\sqrt{3} \simeq 1.7$ also brings it close to 300 $\mathrm{m\,s^{-1}}$, although the adiabatic derivative is the controlled way to calculate the correction. [8]

1.3 One Classical Oscillator in a Heat Bath

We now replace the freely translating molecule by one harmonic oscillator and treat everything else as a heat bath. A mass on a spring is the immediate picture, but the model is much broader. An electromagnetic cavity mode oscillates; atoms in a crystal oscillate; almost any stable system displaced slightly from equilibrium executes harmonic motion to leading order.

With x measured from equilibrium, momentum p, mass m, and spring constant κ, the Hamiltonian is

$$H(x,p) = \frac{p^2}{2m} + \frac{\kappa x^2}{2}, \qquad \omega = \sqrt{\frac{\kappa}{m}}. \tag{1.7}$$

The bath fixes T, and from this point we again use $k_B = 1$, so $\beta = 1/T$. An oscillator equilibrated with the room has a temperature near 300 K; one immersed in liquid helium has a temperature of only a few kelvins. A putative spring placed in the center of the Sun would not survive as a spring, but the example makes the role of the environment clear. The Boltzmann weight factorizes:

$$e^{-\beta H} = e^{-\beta p^2/(2m)} e^{-\beta\kappa x^2/2}. \tag{1.8}$$

Unlike the free gas, the position integral does not merely yield the container volume. The restoring potential weights each displacement:

$$Z_{\mathrm{cl}} = \frac{1}{h}\int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dx\, e^{-\beta p^2/(2m)} e^{-\beta\kappa x^2/2}. \tag{1.9}$$

The factor $1/h$ makes a one-dimensional classical partition function dimensionless. It is suppressed in the board calculation because it is independent of β and cannot affect the mean energy.

For the momentum integral, set

$$q = \sqrt{\frac{\beta}{2m}}\, p, \qquad dp = \sqrt{\frac{2m}{\beta}}\, dq. \tag{1.10}$$

For the position integral, set

$$y = \sqrt{\frac{\beta\kappa}{2}}\, x, \qquad dx = \sqrt{\frac{2}{\beta\kappa}}\, dy. \tag{1.11}$$

Since $\int_{-\infty}^{\infty} e^{-u^2}\, du = \sqrt{\pi}$, the unnormalized phase-space integral used on the board is

$$\int dp\, dx\, e^{-\beta H} = \frac{2\pi}{\beta}\sqrt{\frac{m}{\kappa}} = \frac{2\pi}{\beta\omega}. \tag{1.12}$$

[8] Lecture timestamp: 00:16:54–00:17:27.

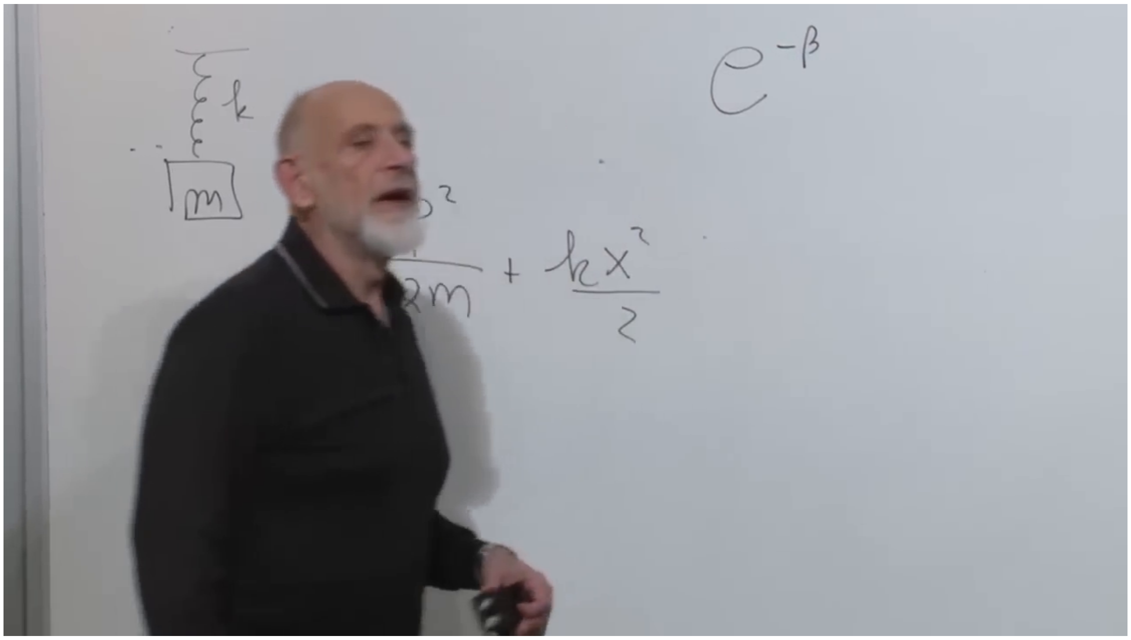

Figure 1.3: A spring-mass oscillator and its kinetic-plus-potential Hamiltonian.
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With the phase-space cell restored,

$$Z_{\mathrm{cl}} = \frac{2\pi}{h\beta\omega} = \frac{1}{\beta\hbar\omega}. \tag{1.13}$$

Only the β-dependence matters for the energy:

$$\begin{aligned} \log Z_{\mathrm{cl}} &= \text{constant} - \log\beta, \\ \langle H\rangle &= -\frac{\partial \log Z_{\mathrm{cl}}}{\partial\beta} \\ &= \frac{1}{\beta} = T. \end{aligned} \tag{1.14}$$

There is no factor of 3 because this is one spatial oscillator. There is also no remaining factor of 1/2, because there are two independent quadratic terms. Each Gaussian contributes $T/2$:

$$\langle K\rangle = \frac{T}{2}, \qquad \langle U\rangle = \frac{T}{2}, \qquad \langle H\rangle = T. \tag{1.15}$$

This is equipartition in its most useful form: every independent quadratic term in the classical Hamiltonian contributes $T/2$.

The answer is independent of both m and κ. The motion itself does depend on those parameters:

$$\left\langle p^2\right\rangle = mT, \qquad \left\langle x^2\right\rangle = \frac{T}{\kappa}, \qquad \left\langle \dot{x}^2\right\rangle = \frac{T}{m}. \tag{1.16}$$

Thus a heavier mass moves more slowly and a stiffer spring has a smaller amplitude, even though the mean energy remains T. That becomes disturbing when κ is made fantastically large. A spring too stiff to move ought to look like a frozen constraint, yet classical equipartition still assigns it energy T.

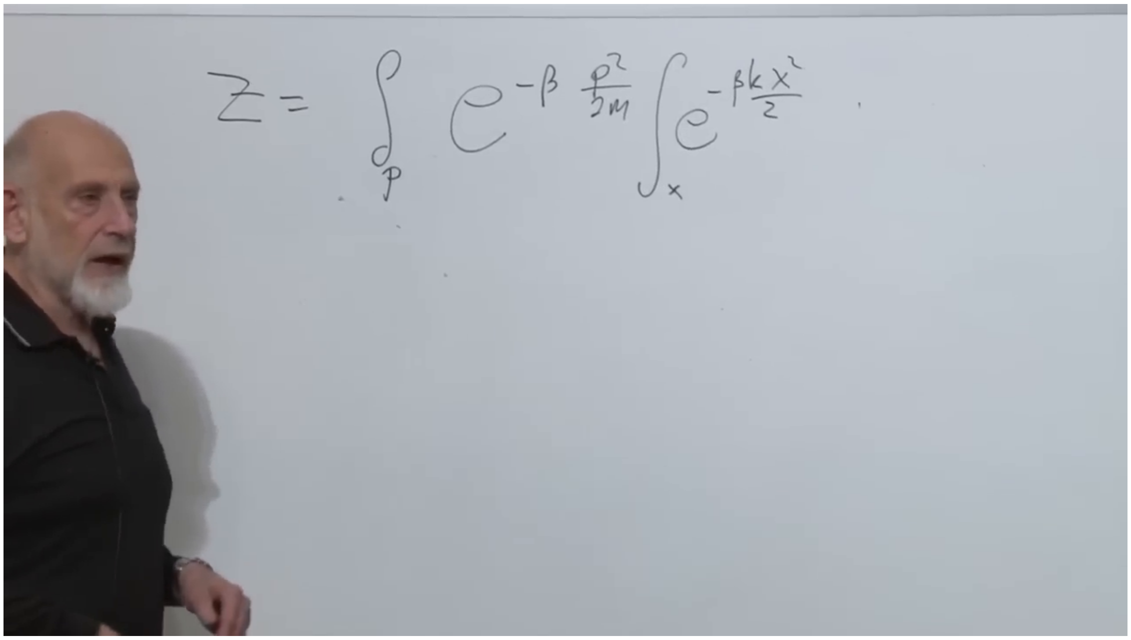

Figure 1.4: The classical oscillator partition function separated into its momentum and displacement Gaussian integrals.

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Question from the audience. As $\kappa \to \infty$, the partition function tends to zero. How can its logarithm still be used?

Response. Keep κ finite but arbitrarily large. Its contribution to $\log Z$ is additive and independent of β, so the β-derivative still gives T. The limiting discomfort is real: the classical calculation is internally consistent, but it omits a feature of nature needed for very stiff oscillators. [9]

Question from the audience. The ideal-gas partition function required an $N!$. Should there be an analogous factorial here?

Response. No. We are studying one specified oscillator, not permuting N identical particles. For many independent labeled modes the partition functions multiply. An indistinguishability factor belongs to a different counting problem. [10]

1.4 Why Classical Equipartition Was a Crisis

Real molecules are not structureless points. A diatomic molecule contains a bond that can vibrate, hence an internal oscillator. A naive classical count would assign that mode additional thermal energy even at ordinary temperatures. In a deliberately schematic count, activating one vibrational oscillator raises an energy of $3T/2$ toward $5T/2$. Precise heat-capacity bookkeeping must also

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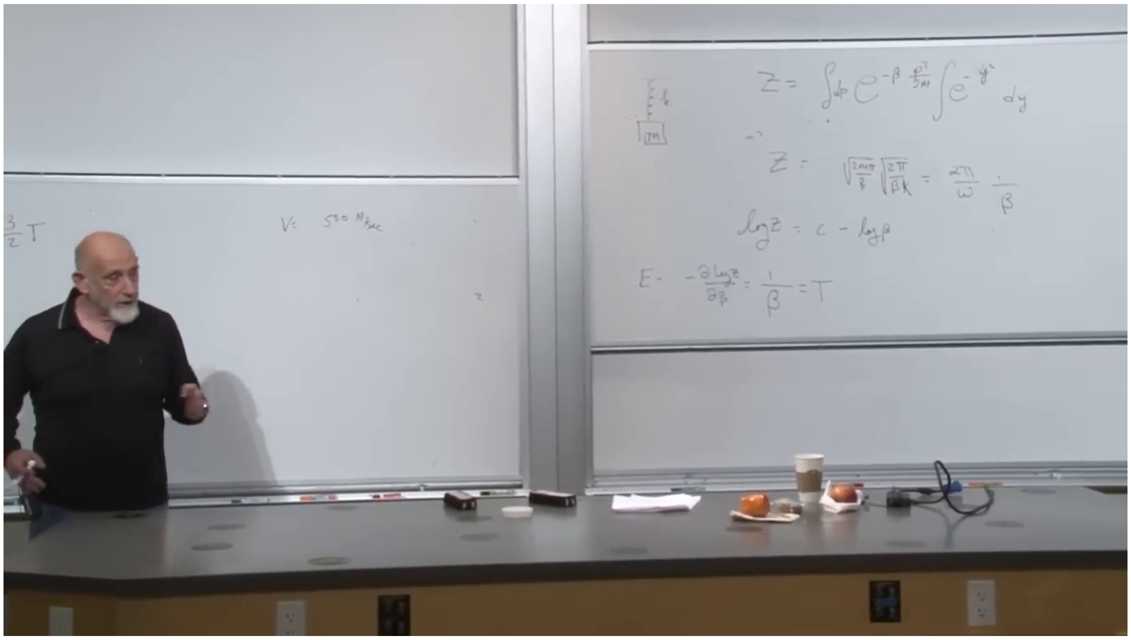

Figure 1.5: The complete classical chain on the right board: Gaussian factorization, $Z \propto (\beta\omega)^{-1}$, and $\langle H \rangle = 1/\beta = T$.

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include rotational degrees of freedom and distinguish energy from its temperature derivative. The decisive experimental fact is simpler: stiff internal modes did not contribute at low temperature as classical equipartition demanded.

One could deny that the molecule has internal structure, which is absurd, or recognize that classical mechanics is missing an ingredient. The ingredient is quantum mechanics. We need very little of it: only that a harmonic oscillator has equally spaced energy levels.

1.5 The Quantum Oscillator

First shift the energy origin and write the excitation spectrum as

$$E_n^{\text{exc}} = n\hbar\omega, \qquad n = 0, 1, 2, \dots. \tag{1.17}$$

The partition function is now a genuine sum rather than a phase-space integral:

$$\begin{aligned} Z_{\text{exc}} &= \sum_{n=0}^{\infty} e^{-\beta n\hbar\omega} \\ &= 1 + x + x^2 + \cdots = \frac{1}{1-x}, \qquad x = e^{-\beta\hbar\omega}, \\ &= \frac{1}{1 - e^{-\beta\hbar\omega}}. \end{aligned} \tag{1.18}$$

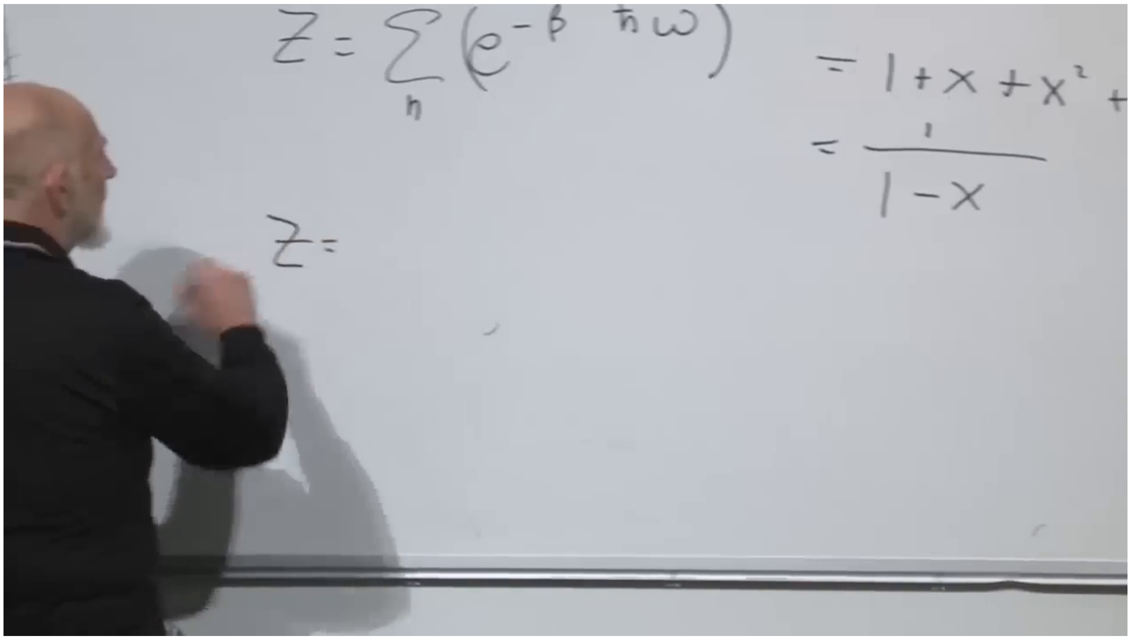

Figure 1.6: Equally spaced levels and the geometric-series evaluation of the quantum oscillator partition function.

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Differentiating gives the mean excitation energy:

$$\begin{aligned} E_{\text{exc}} &= -\frac{\partial \log Z_{\text{exc}}}{\partial \beta} = -\frac{1}{Z_{\text{exc}}}\frac{\partial Z_{\text{exc}}}{\partial \beta} \\ &= \hbar\omega \, \frac{e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} = \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1}. \end{aligned} \tag{1.19}$$

1.5.1 High Temperature

When $\beta\hbar\omega \ll 1$,

$$e^{-\beta\hbar\omega} = 1 - \beta\hbar\omega + O\Big((\beta\hbar\omega)^2\Big). \tag{1.20}$$

The numerator of (1.19) approaches $\hbar\omega$, while the denominator is $\beta\hbar\omega$ to leading order. Therefore

$$E_{\text{exc}} \simeq \frac{1}{\beta} = T. \tag{1.21}$$

The quantum answer recovers classical equipartition when many level spacings fit inside the thermal energy.

1.5.2 Low Temperature

When $\beta\hbar\omega \gg 1$, the exponential is tiny:

$$E_{\text{exc}} \simeq \hbar\omega e^{-\beta\hbar\omega}. \tag{1.22}$$

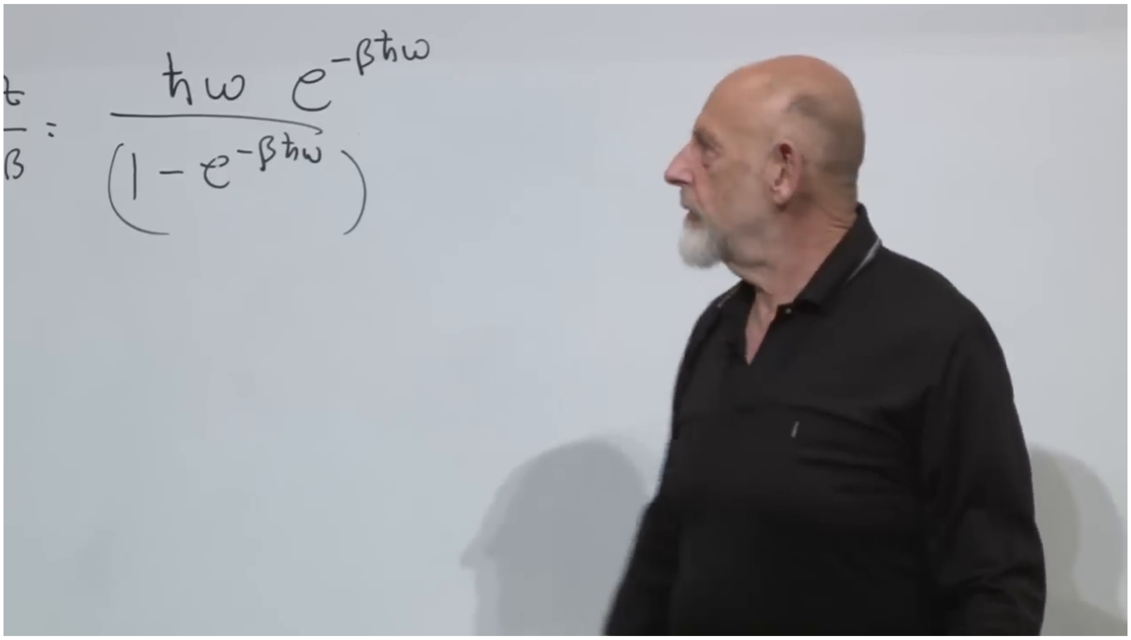

Figure 1.7: The thermal excitation energy obtained from the β-derivative of the quantum partition function.

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The oscillator's thermal excitation is exponentially suppressed. In the shifted convention it is almost zero; the physical zero-point energy will be restored below.

The crossover occurs when

$$\beta\hbar\omega \sim 1, \qquad \text{or equivalently} \qquad T \sim \hbar\omega. \tag{1.23}$$

This condition has a direct meaning. Classical equipartition would give the oscillator energy T. Quantum effects become unavoidable when that scale falls below one level spacing. Thus

$$\begin{aligned} \hbar\omega \gg T: &\quad \text{frozen and quantum,} \\ \hbar\omega \ll T: &\quad \text{active and classical.} \end{aligned} \tag{1.24}$$

A stiffer spring has larger $\omega = \sqrt{\kappa/m}$, so it requires a higher temperature before it behaves classically.

1.6 Degrees of Freedom Appear as We Heat Matter

At low temperature, a diatomic molecule can thermally resemble a point particle because its stiff bond vibration is frozen. As T rises through $\hbar\omega$, the vibrational mode activates. Rotational modes may activate at a lower scale; still higher temperatures reveal more internal motion and eventually ionize the atom. Heating does not merely put more energy into a fixed list of modes. It reveals degrees of freedom that were invisible at lower temperature.

An imaginary nested molecule sharpens the point. Two atoms are joined by a stiff spring, while one apparent atom is itself a pair joined by an even stiffer spring. At low T, both structures look pointlike. At the first threshold the outer vibration appears; at the second threshold the inner structure appears. The complexity of the system is uncovered one energy scale at a time.

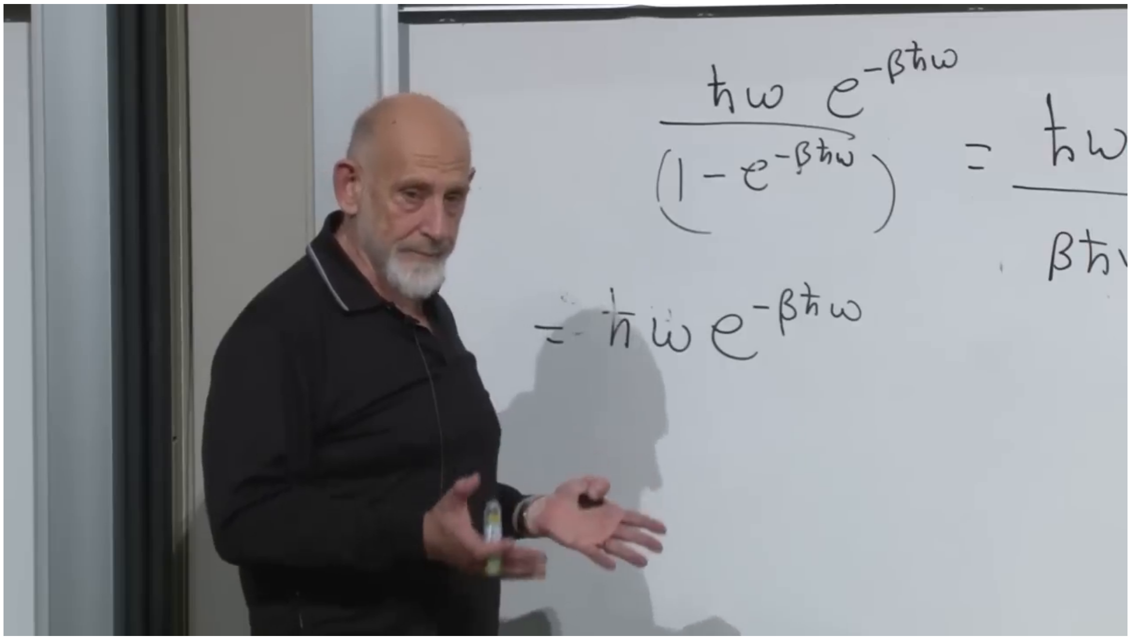

Figure 1.8: The low-temperature expansion: the first thermal excitation is suppressed by $e^{-\beta\hbar\omega}$.
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Question from the audience. Once an oscillator is already in the classical regime, does further heating tighten the spring or change its frequency?

Response. No. The spring constant and frequency remain fixed. Higher temperature produces larger typical momenta and amplitudes, hence larger energy. Stiffness determines the temperature at which the classical law begins, not a temperature-dependent tightening of the spring. [11]

1.6.1 Crystals and Einstein's Specific Heat

A crystal contains a great many vibrational modes. Classical equipartition assigns T-scale energy to every active oscillator and therefore overestimates the low-temperature heat capacity. Einstein's 1907 model treated crystal vibrations quantum mechanically and showed why they are suppressed until thermal energy reaches their level spacing. More accurate solid-state models allow a distribution of mode frequencies, but the freeze-out principle is already present in Einstein's calculation.

Question from the audience. Was this the work for which Einstein received the Nobel Prize, and did it come only after quantum mechanics was established?

Response. No. The Nobel citation concerned the photoelectric effect. The specific-heat model was published in 1907, long before the full foundations of quantum mechanics were understood. Einstein accepted and developed crucial quantum ideas even though he later objected to aspects of the mature theory. [12]

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[12] Lecture timestamp: 00:58:38–01:03:16.

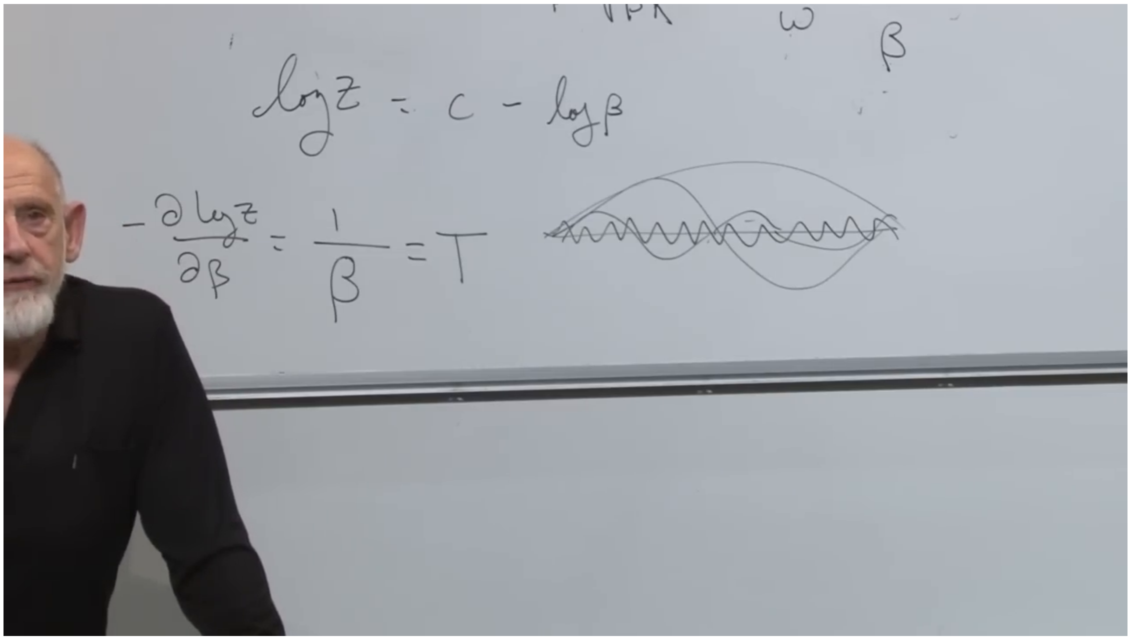

Figure 1.9: Several normal modes superposed on a string, illustrating why a continuous string is an infinite collection of harmonic oscillators.

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Question from the audience. Classical statistical mechanics replaced a sum over discrete states by an apparently uniform integral over position and momentum. What justifies that state counting?

Response. The ultimate justification comes from quantum mechanics. In the high-temperature limit,

$$Z_{\text{exc}} \simeq \frac{1}{\beta\hbar\omega},$$

which exactly matches the dimensionless classical result (1.13). Quantum cells in phase space become the classical phase-space measure when the level spacing is unresolved. [13]

1.6.2 A String Contains Infinitely Many Oscillators

A violin or guitar string decomposes into normal modes: the fundamental, higher harmonics, and progressively shorter wavelengths. Mathematically there are infinitely many oscillator modes. If classical equipartition assigned energy T to every one, the string in thermal equilibrium would have infinite energy. Quantum mechanics prevents that catastrophe. The shortest-wavelength modes have the highest frequencies and remain frozen at any finite temperature; progressively more modes activate as T rises. The same logic will later underlie blackbody radiation.

Question from the audience. Can the quantum-to-classical crossover be observed directly, and can heating reverse the low-temperature behavior?

[13] Lecture timestamp: 00:59:38–01:03:06.

Response. Yes. The temperature dependence of a crystal's specific heat makes the freeze-out and subsequent activation of vibrational modes plainly measurable. [14]

Question from the audience. For many oscillators, do we repeat the single-oscillator partition function for each mode?

Response. For independent modes,

$$Z_{\text{tot}} = \prod_j Z_j, \quad \log Z_{\text{tot}} = \sum_j \log Z_j,$$
$$E_{\text{tot}} = \sum_j E_j.$$

Modes with $\hbar\omega_j \gg T$ make exponentially small thermal contributions; those below the threshold contribute appreciably. [15]

Question from the audience. What happened to the oscillator's zero-point energy $\hbar\omega/2$?

Response. The physical levels are

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega.$$

Consequently

$$Z_{\text{phys}} = e^{-\beta\hbar\omega/2} Z_{\text{exc}},$$
$$E_{\text{phys}} = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1}.$$

The common shift does not change level probabilities, entropy, heat capacity, or thermal excitation energy, but it does shift the absolute mean energy by $\hbar\omega/2$. The multiplying factor in Z is β-dependent, so it must not be discarded when absolute energy is wanted. [16]

1.7 The Second Law as a Reversibility Puzzle

We can now return to the promised problem. The second law says that ordinary macroscopic evolution is irreversible and that entropy tends to increase. Microscopic Hamiltonian evolution is reversible. How can a reversible map of exact states produce an irreversible law of macroscopic states?

Represent the system by a point in a high-dimensional phase space,

$$\Gamma = (q_1, \ldots, q_N, p_1, \ldots, p_N).$$

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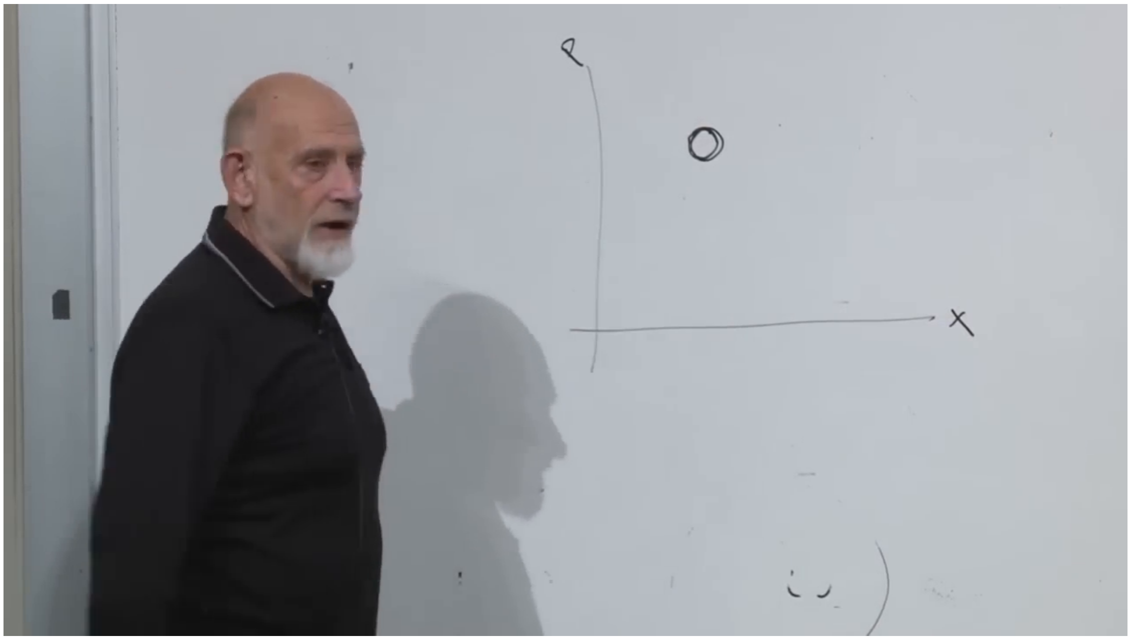

Figure 1.10: A compact probability patch in a schematic two-dimensional slice of the full position-momentum phase space.

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Our knowledge is not usually an exact point. Suppose it is a uniform probability density on a compact patch A of volume Ω_A:

$$f(\Gamma) = \begin{cases} 1/\Omega_A, & \Gamma \in A, \\ 0, & \Gamma \notin A. \end{cases} \tag{1.25}$$

In units with $k_B = 1$, its fine-grained Gibbs entropy is

$$S_{\text{fine}} = -\int d\Gamma\, f \log f = \log \Omega_A. \tag{1.26}$$

A smaller patch means more information and lower entropy. This fixes the sign: increasing uncertainty enlarges the occupied region and raises $\log \Omega_A$.

> **Question from the audience.** Is the system physically constrained to remain inside the drawn patch, and is it already in equilibrium?
>
> **Response.** No. The patch describes what is known about the initial state, not a wall in phase space. It need not be an equilibrium distribution. A gas initially confined to a tank in one corner of a room is a concrete example of special, nonequilibrium information. [17]

1.8 Liouville Flow and Coarse-Grained Growth

Liouville's theorem says that Hamiltonian flow preserves phase-space volume. The exact patch can translate, stretch, fold, and become extremely thin, but

$$\Omega_{A(t)} = \Omega_{A(0)}, \qquad S_{\text{fine}}(t) = S_{\text{fine}}(0). \tag{1.27}$$

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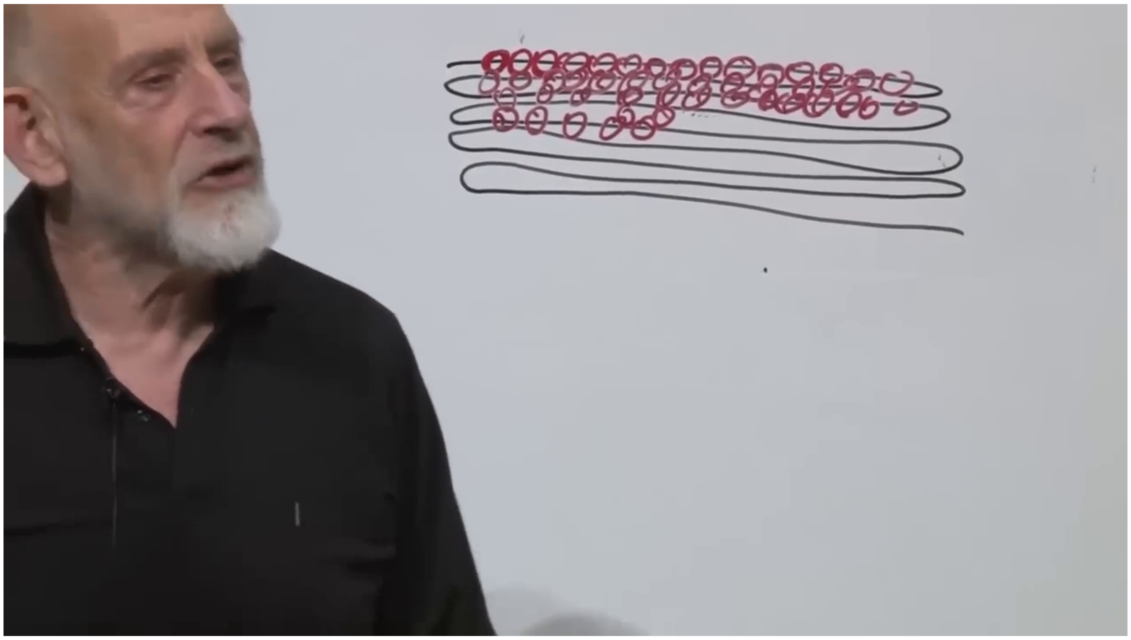

Figure 1.11: A volume-preserving filament viewed through finite-resolution cells. The exact black filament keeps its area; the red coarse cells occupy a larger effective region.

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The discrete analog is a reversible permutation of states. If one initially knows that a die is in one of two states, a one-to-one evolution may change which two states they are, but it does not change the number of possibilities.

Real descriptions have finite resolution. Quantum mechanics supplies a natural phase-space cell, while a classical experiment has finite measuring power even without invoking quantum theory. Under Hamiltonian flow the patch can become a long, thin snake. If its separation between neighboring folds falls below the resolution scale, we replace each unresolved neighborhood by a coarse cell. The exact occupied volume remains fixed, but the union of coarse cells grows:

$$\Omega_{\text{coarse}}(t) \geq \Omega_{A(t)}, \qquad S_{\text{coarse}}(t) = \log \Omega_{\text{coarse}}(t). \tag{1.28}$$

Question from the audience. The coarse cells overlap. Does that violate Liouville's theorem?

Response. No. Liouville governs the exact filament, which remains one-to-one and volume preserving. Overlap belongs to our blurred description: we have discarded information about which thin strand contains the state. [18]

A useful analogy is cotton. The material volume of all fibers can remain nearly fixed while a teased-out wad occupies a large apparent volume to an observer unable to resolve individual fibers. Likewise, a phase-space patch grows finer tentacles; those tentacles develop still finer tentacles. Any fixed finite resolution eventually fails to follow them.

[18] Lecture timestamp: 01:21:32–01:21:50.

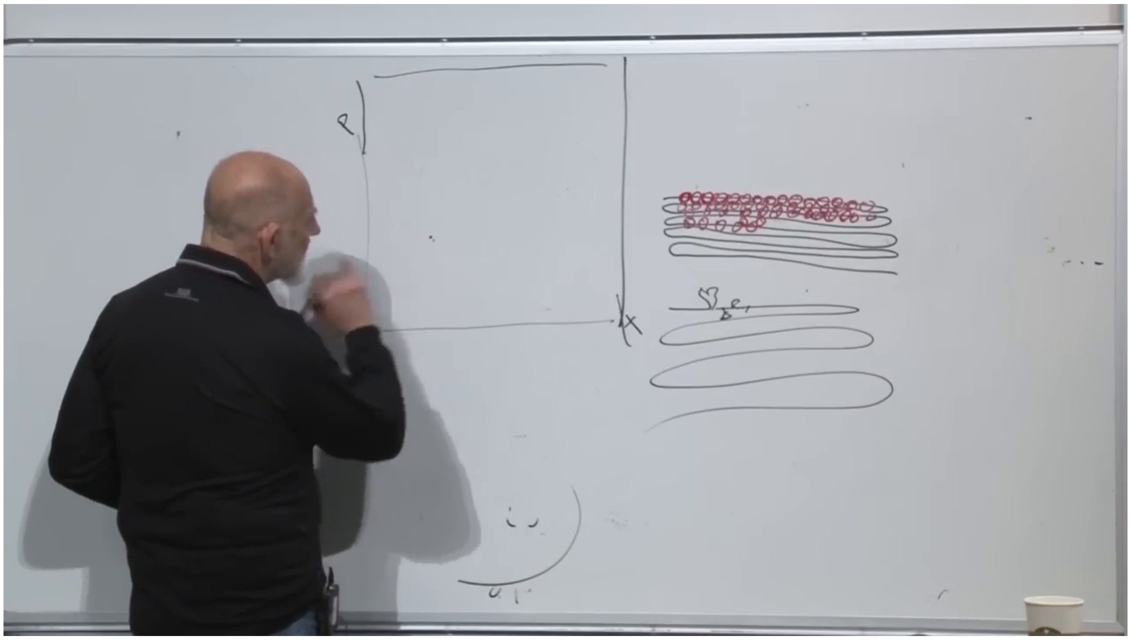

Figure 1.12: A bounded accessible phase-space region beside the filament and its coarse-grained representation.

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Question from the audience. Does entropy merely appear to increase classically, while quantum mechanics makes the increase real by imposing a fundamental resolution limit?

Response. The coarse-graining argument does not require a quantum limit. Entropy refers jointly to the system and the macroscopic distinctions retained in its description. With infinite microscopic information the fine-grained entropy is constant; with any fixed finite resolution, chaotic filaments eventually become unresolved and the coarse-grained entropy increases. [19]

1.9 Chaos, a Finite Accessible Region, and Equilibration

For a gas in a box, positions are bounded. At fixed finite total energy, momenta are bounded as well, even if all energy momentarily concentrates in one particle. The dynamically accessible energy shell therefore has finite measure under the usual confinement assumptions.

Nearby initial points in a chaotic system follow similar histories for a while and then separate. In a billiard break, an imperceptible change in the cue-ball direction is copied into later collisions; the differences compound until the final configurations are unrelated. A whole initial patch is therefore stretched and folded through the accessible region. At finite resolution it approaches a broad, nearly uniform equilibrium description.

Two qualifications matter. Chaos by itself does not prove that every trajectory visits every accessible neighborhood; that stronger sampling claim requires an ergodic or mixing property. Nor is every physical system chaotic. A generic many-body gas and idealized billiards provide the intended setting, where mixing is a useful physical expectation.

[19] Lecture timestamp: 01:24:04–01:25:20.

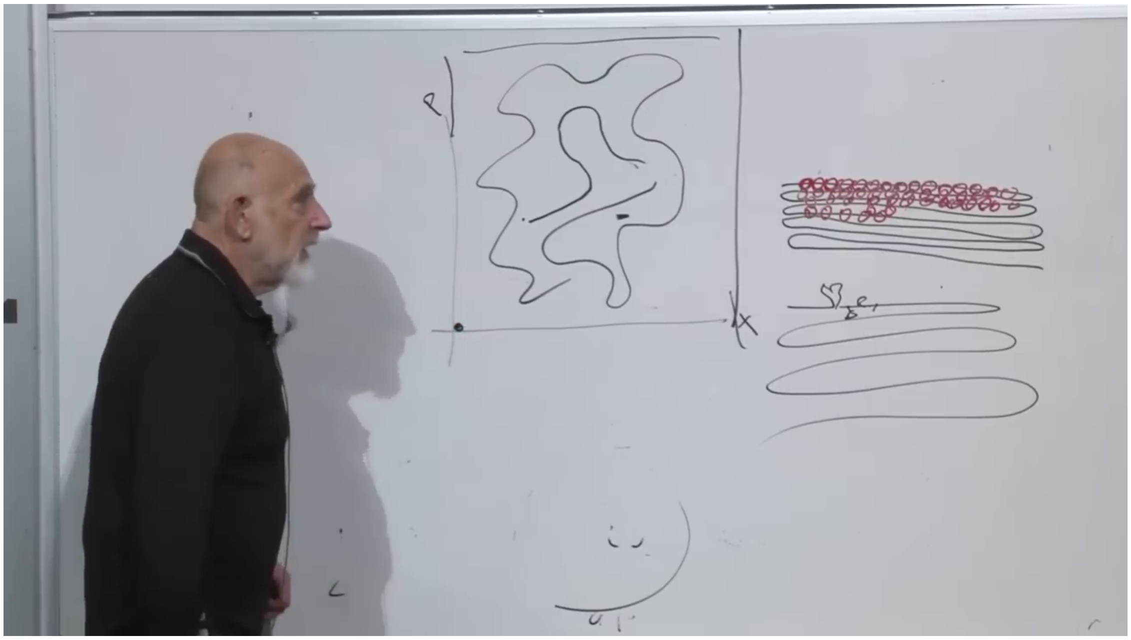

Figure 1.13: A compact special state evolved into a complicated filament through the bounded accessible region while the coarse-grained cells spread over a much larger area.

Lecture frame: 01:33:10.

1.10 Recurrence Does Not Abolish Irreversibility

Suppose all air molecules begin in one corner with a specially prepared velocity pattern. Collisions quickly spread them through the room, and the coarse-grained description reaches equilibrium. In a finite measure-preserving phase space, the Poincare recurrence theorem says that almost every initial point returns arbitrarily close to its starting point. For a tolerance $\epsilon > 0$, a recurrence time $t_R(\epsilon)$ is finite under the theorem's hypotheses, though for a macroscopic system it is fantastically large. Exact return to an infinitely precise point is neither required nor generally expected.

Question from the audience. In the billiard example, does striking the cue ball inject energy and spoil the closed evolution?

Response. The chosen initial state begins immediately after the strike, with that energy already included. Recurrence concerns the subsequent isolated idealized billiard dynamics; a return near the initial state includes a configuration equivalent to the cue ball approaching the rack within the chosen tolerance. [20]

Question from the audience. If the gas can eventually return to the corner, have we lost information, and is the system really reversible?

Response. The microscopic dynamics remains reversible throughout. Coarse-grained entropy can rise because we stop resolving fine correlations, yet an exceptionally rare fluctuation can return the system near its special initial macrostate. Recurrence and coarse-grained irreversibility are therefore compatible. [21]

[20] Lecture timestamp: 01:35:37–01:35:55.

[21] Lecture timestamp: 01:37:34–01:38:15.

This leads to Boltzmann's essential formulation. Entropy does not increase at every logically possible instant. Given a nonequilibrium macrostate of a large system, overwhelmingly more compatible microscopic continuations lead toward larger macrostates than toward still more special ones. Entropy *probably* increases. Downward fluctuations and recurrences are allowed, but their probabilities and time scales make them invisible in ordinary macroscopic experience. The reversibility problem was one of the great historical struggles of statistical mechanics and weighed heavily on Boltzmann. That historical connection should not be turned into a simple causal claim about his life.

1.11 What Chaos Means

For a more operational definition of chaos, suppose two initial phase points differ by $\delta\Gamma(0)$, a chaotic direction typically has

$$\|\delta\Gamma(t)\| \sim e^{\lambda t}\|\delta\Gamma(0)\|, \qquad \lambda > 0, \tag{1.29}$$

over the regime where linearized separation is meaningful. The number λ is a Lyapunov exponent. To predict a time t, the required precision in initial data scales roughly as $e^{-\lambda t}$. Precision in the dynamical law matters too: model errors are amplified along with errors in initial conditions. Weather is the familiar example.

Question from the audience. Is there a principle saying that *most* systems are chaotic, or is that merely empirical?

Response. There are infinitely many chaotic and nonchaotic Hamiltonians, and even the word *most* requires a measure on the space of Hamiltonians. Complicated interacting systems are often chaotic, but there is no elementary universal test that classifies every Hamiltonian by inspection. [22]

Question from the audience. What precisely does *chaotic* mean in this discussion?

Response. Nearby phase trajectories separate rapidly enough that long-time prediction requires exponentially improving knowledge of initial conditions. A harmonic oscillator is the contrast: neighboring phase points rotate together and remain close. The exact two-body Kepler problem is also nonchaotic, while the general three-body problem has chaotic regimes. [23]

Question from the audience. Must the laws of motion, as well as the initial conditions, be known to high precision?

Response. Yes. Both sources of error propagate through the dynamics, and chaotic evolution amplifies both. [24]

The double pendulum supplies a vivid example. An ordinary pendulum is integrable and predictable. Add a second pendulum, and trajectories that begin almost together can eventually transfer energy differently and head in entirely different directions.

[22] Lecture timestamp: 01:39:27–01:40:49.
[23] Lecture timestamp: 01:40:50–01:43:45.
[24] Lecture timestamp: 01:43:46–01:43:55.

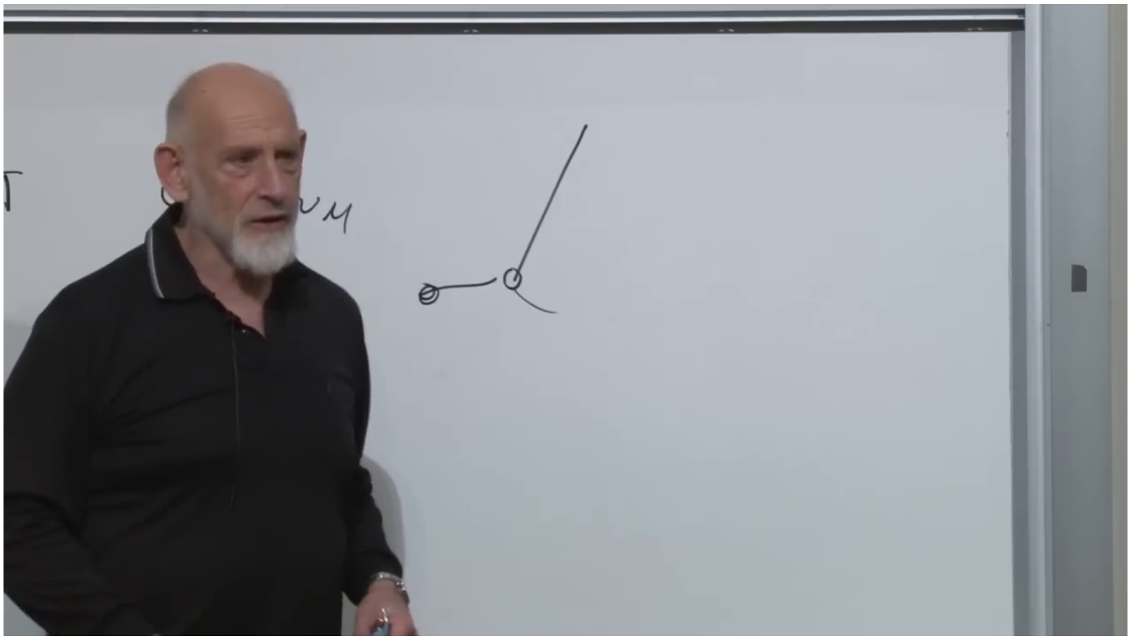

Figure 1.14: The double pendulum as a concrete example of deterministic chaos.
Lecture frame: 01:44:40.

Question from the audience. Why can one mechanical system be chaotic while another is not?

Response. That is a property of the Hamiltonian and its invariant structures, not a consequence of friction or experimental noise. Determining it can be mathematically difficult; complexity is suggestive but not a proof. [25]

Question from the audience. Is an ideal double pendulum still chaotic with no friction?

Response. Yes. Dissipation is not required. Exact deterministic Hamiltonian motion can have sensitive dependence on initial conditions. Infinite-precision initial data would determine the trajectory indefinitely, but finite precision gives a finite prediction horizon. [26]

Question from the audience. Is chaos the same as having no closed-form solution?

Response. No. Chaotic systems generally lack a globally useful closed-form trajectory, but analytic solvability and sensitive dependence are different properties. Chaos is diagnosed dynamically by trajectory separation, not merely by our inability to write a formula. [27]

[25] Lecture timestamp: 01:44:03–01:45:58.
[26] Lecture timestamp: 01:45:57–01:47:54.
[27] Lecture timestamp: 01:48:20–01:49:23.

Question from the audience. How is pervasive chaos different from balancing exactly at the unstable top of a hill?

Response. The hilltop is one exceptional unstable state: away from it, nearby trajectories need not separate rapidly. In a chaotic region, instability is typical along the orbit and nearby phase points repeatedly separate. A positive Lyapunov exponent captures that persistent growth. [28]

1.12 From Microscopic Motion to Statistical Law

The speed-of-sound estimate showed how microscopic thermal motion fixes a macroscopic propagation scale, while the questions exposed the thermodynamic derivative that must be specified. The oscillator then showed the reach and the failure of classical equipartition. Quantum level spacing freezes stiff modes, explains why heat capacities change with temperature, and prevents infinitely many high-frequency modes from carrying infinite thermal energy.

The same progression from exact microphysics to effective macroscopic description resolves the apparent contradiction at the heart of the second law. Hamiltonian flow preserves exact phase-space volume and remains reversible. Chaotic stretching creates finer correlations than a macroscopic description retains, so the coarse-grained occupied region grows. Recurrence guarantees that rare returns are not forbidden; probability explains why they do not define ordinary experience. The second law is not a new irreversible microscopic force. It is the overwhelmingly reliable statistical direction of evolution for special macroscopic initial conditions in systems with enormously many degrees of freedom.

[28] Lecture timestamp: 01:49:27–01:50:11.